Derivatives

Original CC0 evaluation fixture - generated locally

The derivative measures instantaneous rate of change.

For the power function f(x)=x^n, the power rule gives f'(x)=n x^(n-1).

Integrals

Original CC0 evaluation fixture - generated locally

The definite integral represents signed area and accumulation.

The fundamental theorem of calculus connects differentiation and integration: if F'=f, t hen the integral of f from a to b is F(b)-F(a).